ASDA

Sales Performance in Europe

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Introduction

This report presents a comprehensive analysis of ASDA's sales performance across European markets over the period 2010 to 2017. The study aims to evaluate key business metrics, including revenue, profit margins, product performance, and country-wise contributions, in order to gain a deeper understanding of the company's operational and financial trends.

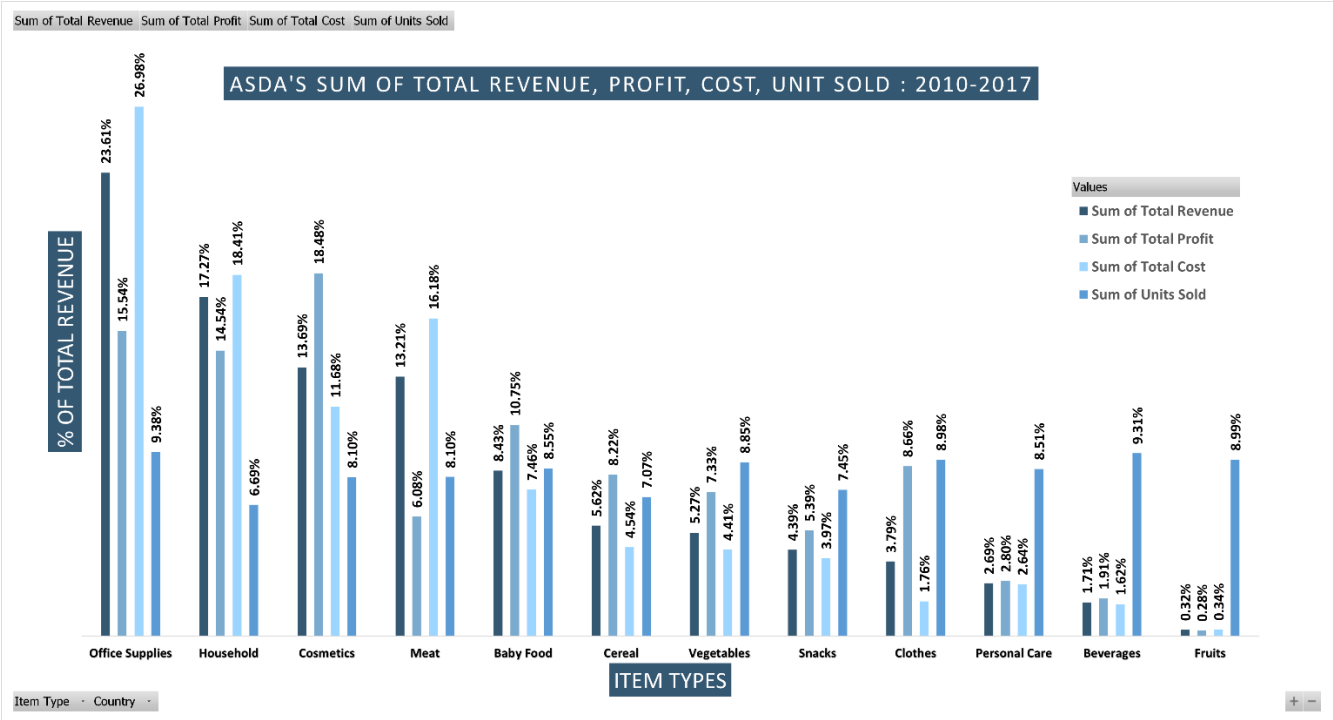
The analysis begins with an exploration of overall sales performance, highlighting key revenue drivers, high-performing product categories, and underperforming segments. Monthly and yearly revenue patterns are examined to identify seasonal trends and fluctuations in sales. This is followed by a detailed time series decomposition, which breaks down the data into trend, seasonality, and residual components to better understand the underlying structure of the sales data.

Given the variability and irregular patterns observed across several countries, the analysis focuses on the top 10 revenue-generating countries. This allows for more reliable insights and meaningful forecasting, as these markets exhibit relatively consistent sales behaviour compared to smaller or more volatile regions.

Furthermore, forecasting techniques are applied to predict future sales performance, providing insights into potential growth opportunities and risks. Country-wise analysis is conducted to evaluate market stability, identify high-potential regions, and support strategic decision-making.

Overall, this report aims to combine statistical analysis with business interpretation to provide actionable insights that can support ASDA's strategic planning, resource allocation, and market expansion across Europe.

Sales Analysis



Monthly Sales

March, June, Jan seems to be top revenue generating month, contributing 494.67 million of sales altogether followed by May & July respectively.

October, April, August seems to be low revenue generating month respectively contributing 339.84 million of sales in seven years altogether. (See Month Generating High Revenue 2010-2017 Table)

Row Labels	Sum of Total Revenue	Sum of Total Profit	Sum of Total Cost	Sum of Units Sold
Office Supplies	40,22,13,995.61	7,79,77,176.25	32,42,36,819.36	6,17,641.00
Household	29,42,05,199.23	7,29,62,466.77	22,12,42,732.46	4,40,249.00
Cosmetics	23,31,54,825.20	9,27,23,306.17	14,04,31,519.03	5,33,291.00
Meat	22,50,26,000.64	3,05,09,107.20	19,45,16,893.44	5,33,376.00
Baby Food	14,36,47,587.68	5,39,40,997.16	8,97,06,590.52	5,62,706.00
Cereal	9,57,91,404.50	4,12,55,034.15	5,45,36,370.35	4,65,685.00
Vegetables	8,97,46,728.64	3,67,76,002.72	5,29,70,725.92	5,82,544.00
Snacks	7,47,88,612.80	2,70,27,422.40	4,77,61,190.40	4,90,160.00
Clothes	6,46,26,552.80	4,34,31,314.40	2,11,95,238.40	5,91,385.00
Personal Care	4,58,08,030.40	1,40,45,628.80	3,17,62,401.60	5,60,480.00
Beverages	2,90,93,160.85	96,01,662.78	1,94,91,498.07	6,13,133.00
Fruits	55,20,299.76	14,25,929.52	40,94,370.24	5,91,672.00
Grand Total	**1,70,36,22,398.11**	**50,16,76,048.32**	**1,20,19,46,349.79**	**65,82,322.00**

Total Revenue

Top 3 Key Revenue Drivers

1. **Office Supplies** generate most of the revenue contributing **23.61% (402 M)** of Total Sales of which year 2015` was the **highest 18.76%** and 2017 being the **lowest 8.51%** of total revenue. **Slovakia, Kosovo, Ukraine, Russia are the top purchaser** of office supplies respectively. Vatican City, Monaco, United Kingdom, Switzerland is the lowest purchaser generating approx. around 0.07% of the Total Revenue
2. **Household** generate second most of the revenue contributing **17.27% (294 M)** of Total Sales of which the year 2015, 2016 was the **highest approx. 15%** and 2017 being the **lowest 7.54%** of total revenue. **Czech Republic, Andorra, Bosnia and Herzegovina, Macedonia are the top purchaser** of household respectively. Spain, Moldova, Switzerland, Belgium is the lowest purchaser generating approx. around 0.15% of the Total Revenue.
3. **Cosmetics** generate third most of the revenue contributing **13.69% (233 M)** of Total Sales of which the year 2012 was the **highest approx. 24.30%** and 2013 being the **lowest 6.22%** of total revenue. **Malta, Ukraine, Albania, Switzerland, Andorra are the top purchaser** of cosmetics respectively. Vatican City, Monaco, Cyprus, Macedonia, Liechtenstein is the lowest purchaser generating approx. around 0.17% of the Total Revenue.

Underperforming products

1. **Fruits** are least revenue generating items contributing **0.32% (5 M)** of total sales mostly purchased by **Greece, Kosovo, Slovenia, Belgium, Bulgaria**. Year **2016** was the most sales made **approx. 18.13%** of total Fruits Sales made from 2010 till 2017
2. Beverages are second least revenue generating items contributing 1.71% (29 M) of total sales mostly purchased by Macedonia, Belarus, Andorra, Serbia, Slovakia, Estonia, Netherlands, Austria, Greece, Moldova, Ukraine, Hungary, Slovenia, Finland, San Marino making 1% of the total revenue. Year 2015,2016 was the most sales made approx. 13% of total Beverage's sales made from 2010 till 2017
3. **Personal Care** are the third lowest revenue generating items contributing **2.69% (45 M)** of total sales mostly purchased by **France, United Kingdom, Latvia, San Marino, Cyprus, Romania, Ireland, Ukraine making 1%** of the total revenue. Year **2014** was the most sales made **approx. 19%** of total Personal's care sales made from 2010 till 2017

Product-wise Sales and Profitability Analysis

Overview

- This section evaluates the performance of different product categories based on units sold, total revenue, cost structure, and profit margins. The objective is to identify high-performing products, profitability patterns, and areas that require strategic improvement.

Revenue and Sales Performance

- Among all product categories, **Office Supplies** generates the highest revenue, contributing over 40 million in total sales. This is followed by **Household** and **Cosmetics**, which also represent significant portions of overall revenue. These categories are key drivers of ASDA's sales performance.
- In terms of units sold, **Beverages**, **Office Supplies**, and **Fruits** show high sales volume, indicating strong demand. However, high volume does not always translate into high profitability.

Profitability Analysis

- **Clothes** stand out as the most profitable category, with a profit margin of **67.20%**, followed by **Cereal (43.07%)** and **Vegetables (40.98%)**. These categories demonstrate efficient cost management and high return on sales.
- On the other hand, **Meat (13.56%)**, **Office Supplies (19.39%)**, and **Household (24.80%)** have relatively low profit margins despite generating high revenue. This suggests that these products operate with higher costs and lower efficiency.

Cost Structure Insights

- The cost percentage varies significantly across products:
- **Meat (86.44%)** and **Office Supplies (80.61%)** have the highest cost proportions, reducing profitability
- **Clothes (32.80%)** has the lowest cost percentage, contributing to its high profit margin
- Most other products fall within the 55%–70% cost range
- This indicates that cost control plays a critical role in determining overall profitability.

Key Insights

- High revenue does not always indicate high profitability
- **Clothes, Cereal, and Vegetables** are the most efficient in terms of profit generation
- **Office Supplies and Household**, although top revenue contributors, have lower margins due to high costs
- **Meat** is the least profitable category and may require cost optimization
- Some low-revenue items like **Fruits** also show weak profitability, making them less attractive from a business perspective

Row Labels	Sum of Units Sold	Sum of Total Cost	Sum of Total Profit	Sum of Total Revenue	%Cost	%Profit
Clothes	5,91,385.00	2,11,95,238.40	4,34,31,314.40	6,46,26,552.80	32.80%	67.20%
Cereal	4,65,685.00	5,45,36,370.35	4,12,55,034.15	9,57,91,404.50	56.93%	43.07%
Vegetables	5,82,544.00	5,29,70,725.92	3,67,76,002.72	8,97,46,728.64	59.02%	40.98%
Cosmetics	5,33,291.00	14,04,31,519.03	9,27,23,306.17	23,31,54,825.20	60.23%	39.77%
Baby Food	5,62,706.00	8,97,06,590.52	5,39,40,997.16	14,36,47,587.68	62.45%	37.55%
Snacks	4,90,160.00	4,77,61,190.40	2,70,27,422.40	7,47,88,612.80	63.86%	36.14%
Beverages	6,13,133.00	1,94,91,498.07	96,01,662.78	2,90,93,160.85	67.00%	33.00%
Personal Care	5,60,480.00	3,17,62,401.60	1,40,45,628.80	4,58,08,030.40	69.34%	30.66%
Fruits	5,91,672.00	40,94,370.24	14,25,929.52	55,20,299.76	74.17%	25.83%
Household	4,40,249.00	22,12,42,732.46	7,29,62,466.77	29,42,05,199.23	75.20%	24.80%
Office Supplies	6,17,641.00	32,42,36,819.36	7,79,77,176.25	40,22,13,995.61	80.61%	19.39%
Meat	5,33,376.00	19,45,16,893.44	3,05,09,107.20	22,50,26,000.64	86.44%	13.56%
Grand Total	6582322	1201946350	501676048.3	1703622398	70.55%	29.45%

Business Implications

- Focus on expanding high-margin products such as **Clothes and Cereal** to improve overall profitability
- Improve cost efficiency in high-revenue categories like **Office Supplies and Household**
- Reevaluate low-performing categories such as **Meat and Fruits** to determine whether cost reduction or strategic repositioning is required
- Align pricing and supply chain strategies to optimize margins across all product categories

Conclusion

- The analysis highlights a clear distinction between revenue generation and profitability. While certain product categories drive overall sales, others contribute more effectively to profit. Therefore, ASDA should adopt a balanced strategy that focuses on both revenue growth and cost efficiency to maximize overall business performance.

Itemise Sale

Office Supplies, Household and Cosmetics contributes 50% sales of the total revenue of which Cosmetics generate the highest profit margin up to 18.48%

Row Labels	Sum of Total Revenue	Sum of Total Profit
Cosmetics	13.69%	18.48%
Office Supplies	23.61%	15.54%
Household	17.27%	14.54%
Baby Food	8.43%	10.75%
Clothes	3.79%	8.66%
Cereal	5.62%	8.22%
Vegetables	5.27%	7.33%
Meat	13.21%	6.08%
Snacks	4.39%	5.39%
Personal Care	2.69%	2.80%
Beverages	1.71%	1.91%
Fruits	0.32%	0.28%
Grand Total	**100.00%**	**100.00%**

Monthly and Yearly Revenue Analysis

Sum of Total Revenue	Column Labels								
Row Labels	2010	2011	2012	2013	2014	2015	2016	2017	Grand Total
Mar	8.5 M	11.2 M	27.8 M	30.4 M	32.1 M	22.0 M	14.6 M	24.1 M	170.7 M
Jun	25.1 M	22.5 M	28.0 M	13.9 M	31.6 M	14.5 M	13.6 M	16.9 M	166.1 M
Jan	21.1 M	21.2 M	34.0 M	6.0 M	24.8 M	15.5 M	18.6 M	16.7 M	157.9 M
May	20.3 M	21.2 M	24.5 M	6.7 M	23.5 M	23.9 M	17.3 M	20.3 M	157.7 M
Jul	12.7 M	25.7 M	13.4 M	12.2 M	21.0 M	14.6 M	32.9 M	23.5 M	155.9 M
Dec	30.7 M	30.6 M	20.6 M	14.3 M	9.6 M	25.1 M	16.3 M		147.1 M
Feb	19.9 M	21.2 M	30.3 M	16.6 M	6.7 M	14.8 M	11.3 M	20.8 M	141.5 M
Nov	6.0 M	16.1 M	23.9 M	31.2 M	11.7 M	34.7 M	15.8 M		139.5 M
Sep	18.8 M	14.5 M	18.0 M	18.8 M	19.4 M	20.4 M	17.4 M		127.3 M
Aug	12.4 M	12.8 M	20.5 M	4.3 M	33.7 M	12.8 M	20.8 M		117.2 M
Apr	10.1 M	17.3 M	26.5 M	14.9 M	13.3 M	12.3 M	6.0 M	14.4 M	114.8 M
Oct	23.9 M	17.5 M	18.1 M	13.7 M	10.3 M	17.3 M	7.0 M		107.8 M
Grand Total	**209.4 M**	**231.9 M**	**285.5 M**	**183.0 M**	**237.8 M**	**228.0 M**	**191.5 M**	**136.6 M**	**1703.6 M**

Overview

This section analyses the monthly and yearly distribution of total revenue for ASDA across the observed period (2010–2017). The objective is to identify key seasonal patterns, high-performing periods, and overall business trends.

Yearly Revenue Performance

The analysis shows that ASDA experienced its highest revenue in 2012 (285.5M), marking a peak in overall business performance. This was followed by a noticeable decline in 2013 (183.0M). Although some recovery is observed in 2014 and 2015, revenue remains inconsistent in later years, with 2017 recording the lowest performance (136.6M).

This trend indicates that the company experienced a strong growth phase initially, followed by a period of instability and fluctuating performance. Such variations may be attributed to market conditions, operational changes, or external economic factors.

Monthly Revenue Patterns

The monthly analysis highlights clear variations in sales performance across different months:

March (170.7M) emerges as the highest-performing month overall, followed closely by June (166.1M).

Other strong months include January, May, and July, all contributing significantly to total revenue.

In contrast, October (107.8M), April (114.8M), and August (117.2M) represent the lowest-performing months.

These findings suggest that sales tend to peak during the early part of the year (Q1) and mid-year periods, while weaker performance is observed in late summer and early autumn.

Seasonality Insights

The data indicates the presence of moderate seasonality in sales:

Revenue generally increases from January to March

A slight decline is observed in April

Sales rise again during May to July

A downward trend appears in August to October

Some recovery is seen towards the end of the year (November–December), although not consistently across all years

However, the seasonal pattern is not perfectly stable, as variations occur across different years. This suggests that while seasonal effects exist, they are influenced by additional factors such as promotions, demand shifts, or external conditions.

Key Insights

The business achieved peak performance in 2012, followed by a decline and fluctuating trend

March and June are consistently the strongest months for revenue generation

October, April, and August represent weaker periods with lower sales

Seasonality is present but not consistent across all years

Revenue patterns indicate a combination of seasonal demand and irregular fluctuations

Business Implications

The findings provide valuable insights for strategic decision-making:

High-performing months such as March and June should be leveraged for maximizing revenue through increased inventory and targeted marketing campaigns

Low-performing months such as October and August require strategic interventions, including promotions and discounts, to stimulate demand

The variability in yearly performance suggests the need for more adaptive and data-driven planning

Conclusion

The analysis demonstrates that ASDA's revenue performance is influenced by both seasonal patterns and irregular fluctuations. While certain months consistently generate higher revenue, the lack of stability across years highlights the importance of flexible strategies. Understanding these patterns can help improve forecasting accuracy and support more effective business planning.

Sales Channel

Offline Sales Performance

The offline channel demonstrates relatively strong and consistent performance across most months:

- **March (6.04%)** is the highest revenue-generating month, followed by **June (5.55%)** and **December (5.05%)**
- Other strong months include **May, November, and February**, all contributing above 4%
- The lowest-performing months are **October (2.68%)** and **April (2.70%)**

In terms of profit:

- Profit distribution closely follows revenue trends

- Slightly higher profit contribution is observed in **January (4.66%)**, indicating better margins during this period

Insight: Offline sales show **stable performance with clear peak months**, especially during early and mid-year periods.

Online Sales Performance

The online channel shows a slightly different pattern:

- **January (5.22%)** is the top-performing month, followed by **July (5.04%)** and **May (4.71%)**
- Other consistent months include **June, April, and February**
- Lower performance is observed in **August (3.11%)** and **September (3.43%)**

In terms of profit:

- **January (5.87%)** generates the highest profit contribution
- Profit margins appear relatively strong in months like **October (4.06%)** despite moderate revenue

Insight: Online sales are **more evenly distributed across months**, with less extreme peaks compared to offline sales.

Row Labels	Sum of Total Revenue	Sum of Total Profit
Offline		
Mar	6.04%	5.82%
Jun	5.55%	5.52%
Dec	5.05%	4.41%
May	4.55%	4.74%
Nov	4.43%	4.10%
Feb	4.30%	4.36%
Jul	4.11%	4.10%
Sep	4.05%	4.28%
Jan	4.04%	4.66%
Aug	3.77%	3.50%
Apr	2.70%	2.44%
Oct	2.68%	2.65%
Online		
Jan	5.22%	5.87%
Jul	5.04%	4.97%
May	4.71%	4.45%
Jun	4.20%	4.49%
Apr	4.04%	4.28%
Feb	4.00%	3.52%
Mar	3.98%	4.31%
Nov	3.76%	3.46%
Oct	3.65%	4.06%
Dec	3.58%	3.22%
Sep	3.43%	3.76%
Aug	3.11%	3.03%
Grand Total	**100.00%**	**100.00%**

Key Differences Between Online and Offline Channels

- Offline channel peaks in March and June, while Online channel peaks in January and July
- Offline sales show stronger seasonal spikes, whereas online sales are more balanced throughout the year

- Profit margins in the online channel are relatively strong in certain months, even when revenue is moderate
- Offline sales rely more on **traditional seasonal demand**, while online sales reflect **consistent consumer activity**

Seasonal Insights

- Both channels show strong performance during **early-year months (January–March)**
- Mid-year (May–July) also contributes significantly to revenue
- Late summer (August–September) tends to be weaker across both channels
- End-of-year (November–December) performance is stronger in offline than online

Business Implications

Offline Strategy: Focus on maximizing sales during peak months such as March and June through inventory planning and promotional campaigns

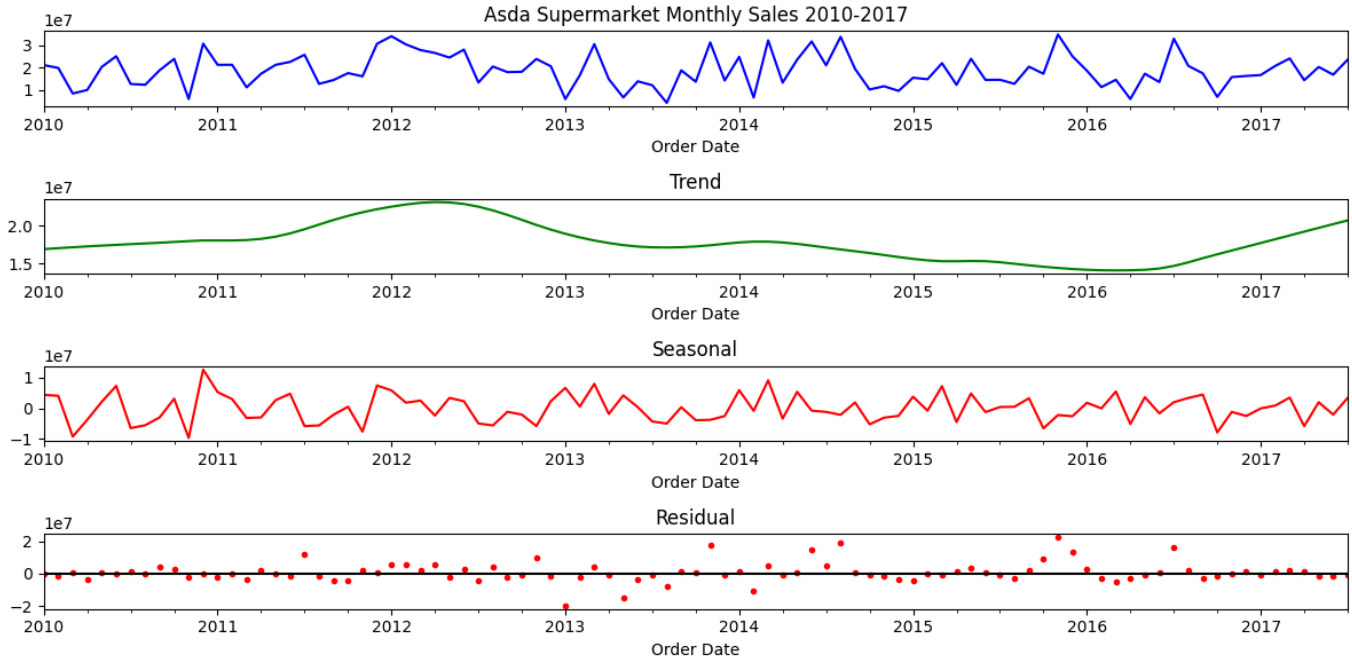
Online Strategy: Maintain consistent engagement throughout the year, with targeted campaigns in January and mid-year periods

Low-performing months (August–October): Introduce discounts, marketing campaigns, and digital promotions to boost demand

Conclusion

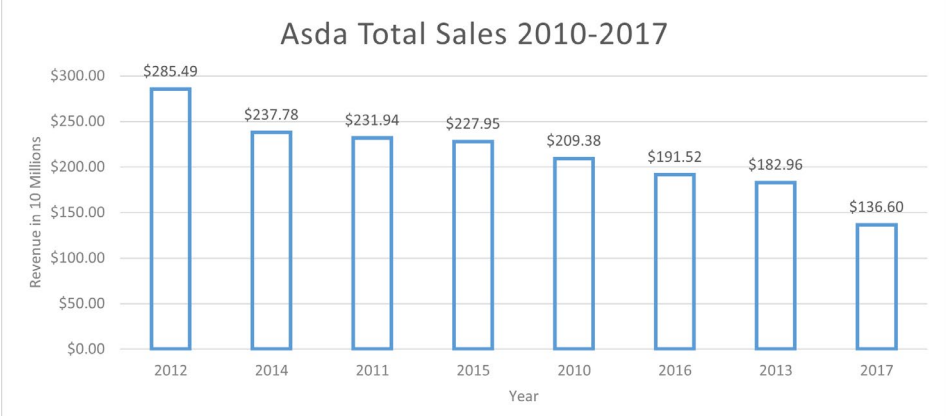
The comparison highlights that while offline sales are more influenced by seasonal peaks, online sales provide a more stable and consistent revenue stream. A balanced strategy that leverages the strengths of both channels can help ASDA optimize overall performance and improve profitability.

Time Series Decomposition Insights – Europe Sales (2010–2017)



Observed Data

The observed time series shows significant fluctuations in monthly sales across the entire period. Sales exhibit frequent spikes and drop, indicating high short-term variability. Despite this volatility, an underlying structure is visible when analysed further through decomposition.



Trend Analysis

The trend component reveals a **non-linear pattern**:

- Growth phase in earlier years
- Period of decline or stabilization
- Recovery phase in later years

This indicates that sales performance is not constant and may be affected by business cycles or external economic factors.

Seasonality Analysis

Seasonality is present but not strongly pronounced:

- Some recurring monthly patterns exist
- However, the magnitude of seasonal effects varies over time
- This suggests **weak to moderate seasonality**, likely influenced by holidays or promotional events rather than consistent cyclical behaviour.

Residual Analysis

The residual component shows:

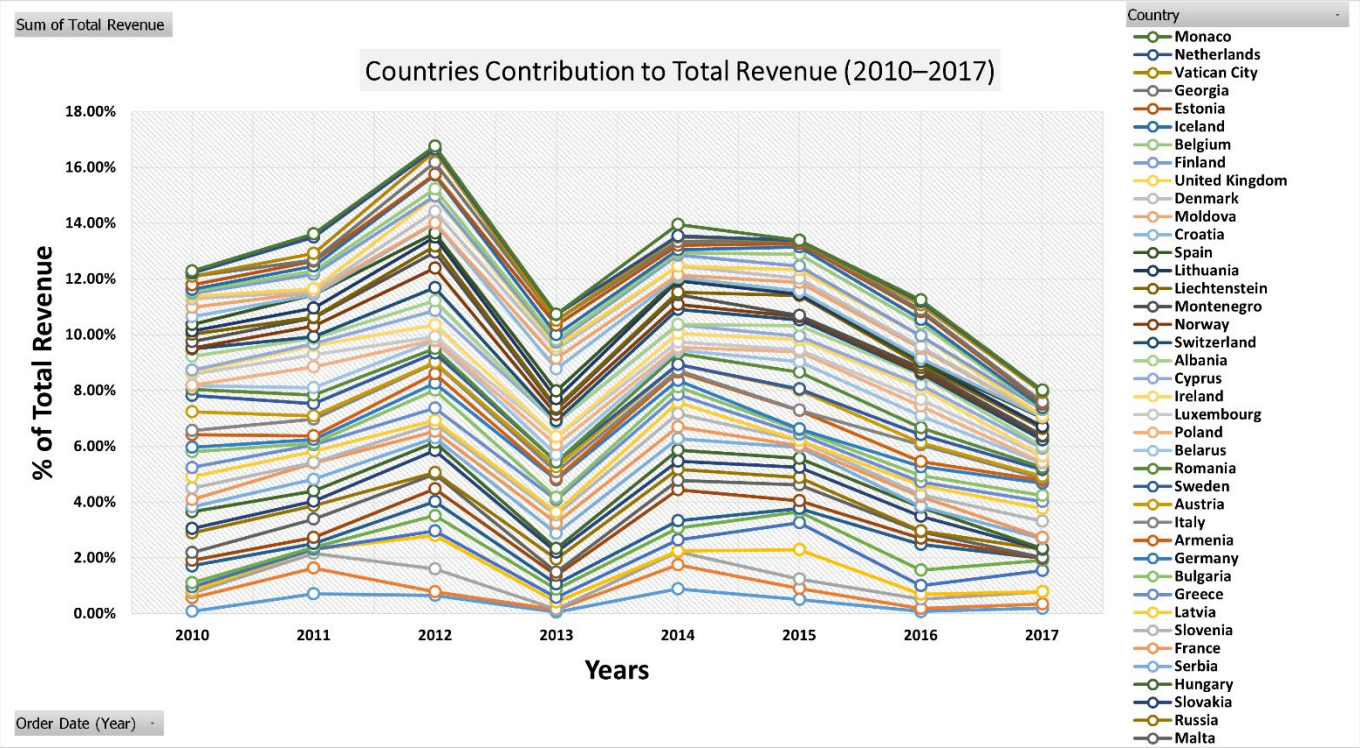
- Large and irregular spikes
- High variability not explained by trend or seasonality
- This indicates the presence of **significant noise** in the data, which reduces forecasting accuracy and suggests that additional external factors influence sales.

Conclusion

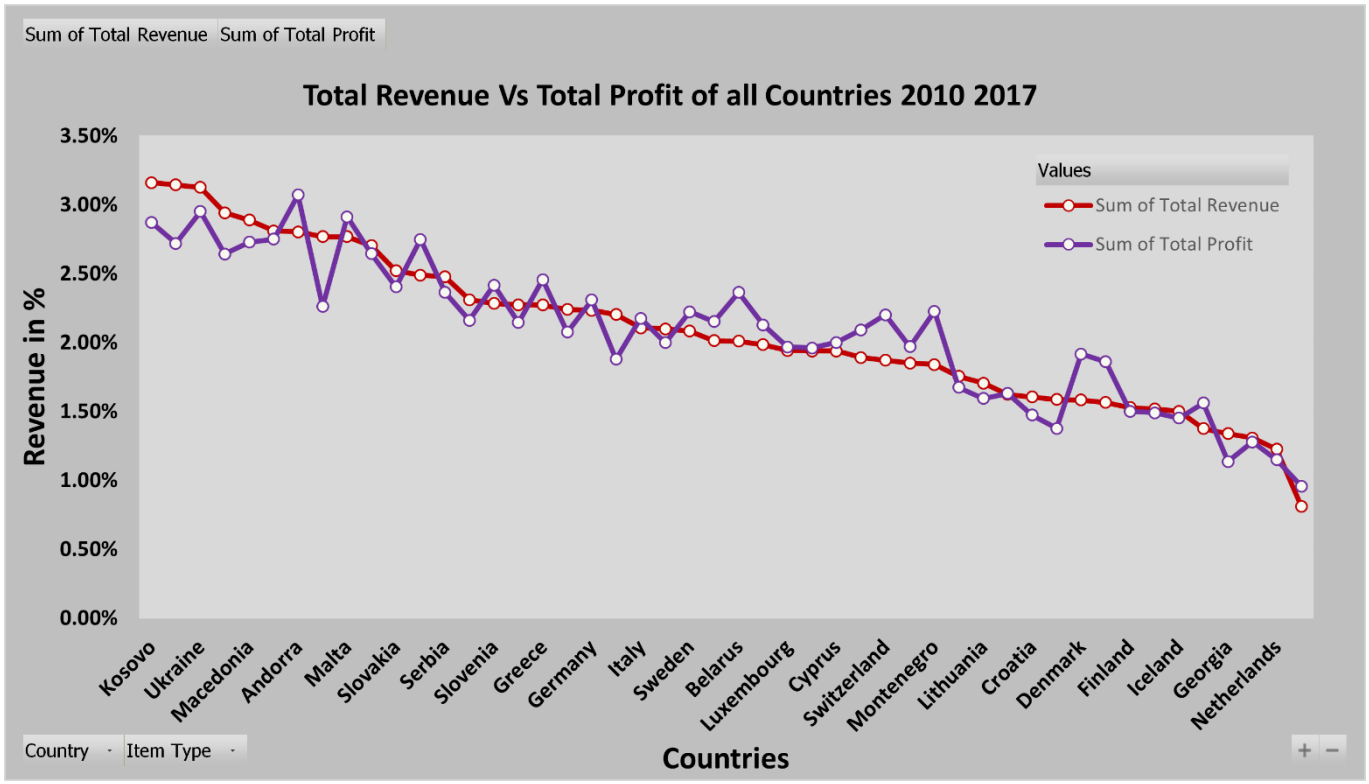
Due to the high variability and sparsity observed in several countries with low revenue, analysing the entire dataset did not provide clear or reliable patterns for time series modelling. Therefore, the analysis was restricted to the top revenue-generating countries. These countries exhibit more consistent sales behaviour, allowing for more accurate identification of trends, seasonality, and improved forecasting performance. However, this approach focuses on major contributors and may not fully represent smaller markets.

Countries Analysis

Top Ten Country-wise Contribution to Total Revenue (2010–2017)



- **Kosovo** contributed approximately **3.16%** to the **total revenue** during the period 2010–2017, generating around **53.83 million.**
- The **Czech Republic** accounted for **approximately 3.14%** of **total revenue** between 2010 and 2017, with revenue of about **53.54 million.**
- **Ukraine** contributed **nearly 3.13%** to the **overall revenue**, amounting to approximately **53.25 million** during the same period.
- **Bosnia and Herzegovina** generated **around 2.94%** of **total revenue,** with an approximate value of **50.12 million.**
- **Macedonia** accounted for **approximately 2.89%** of **total revenue**, contributing around **49.22 million**.
- **San Marino** contributed close to **2.81%** of the **total revenue**, generating approximately **47.88 million.**
- **Andorra** accounted for about **2.80%** of **total revenue**, with a contribution of approximately **47.76 million.**
- **Portugal** contributed nearly **2.77%** to **total revenue**, amounting to around **47.17 million**.
- **Malta** also represented **approximately 2.77%** of **total revenue**, generating about **47.15 million.**
- **Russia** contributed **around 2.70%** of the **overall revenue**, with an approximate value of **46.05 million**.



- **Balanced Revenue Distribution Across Mid-Tier Countries**

  The revenue contribution is relatively evenly distributed among these countries, with all shares falling within a narrow range of 2.70% to 3.16%. This indicates a well-diversified revenue base, reducing dependency on any single market and lowering overall business risk.

- **Emerging Markets Show Consistent Contribution**

  Countries like Kosovo, Ukraine, and the Czech Republic are among the top contributors within this group, each generating over 3% of total revenue (~53M). This suggests that emerging or smaller European markets are performing strongly and consistently, highlighting potential opportunities for further expansion and investment.

- **Major Revenue Drop After 2012–2014 Peak**

  Total revenue peaks around 2012 (~285M) and 2014 (~237M)

  Then sharp decline by 2017 (~136M)

  There is a clear downward trend after 2014, suggesting:

  Market saturation

  External shock (economic / policy / competition)

  Weak retention strategy

- **High Volatility in Country Performance (Unstable Markets)**

  Countries like: Bosnia & Herzegovina, Croatia, Poland, Austria, show huge spikes + sudden drops. Example:

  Bosnia: 20M (2012) → 0.3M (2014) → 17M (2015)

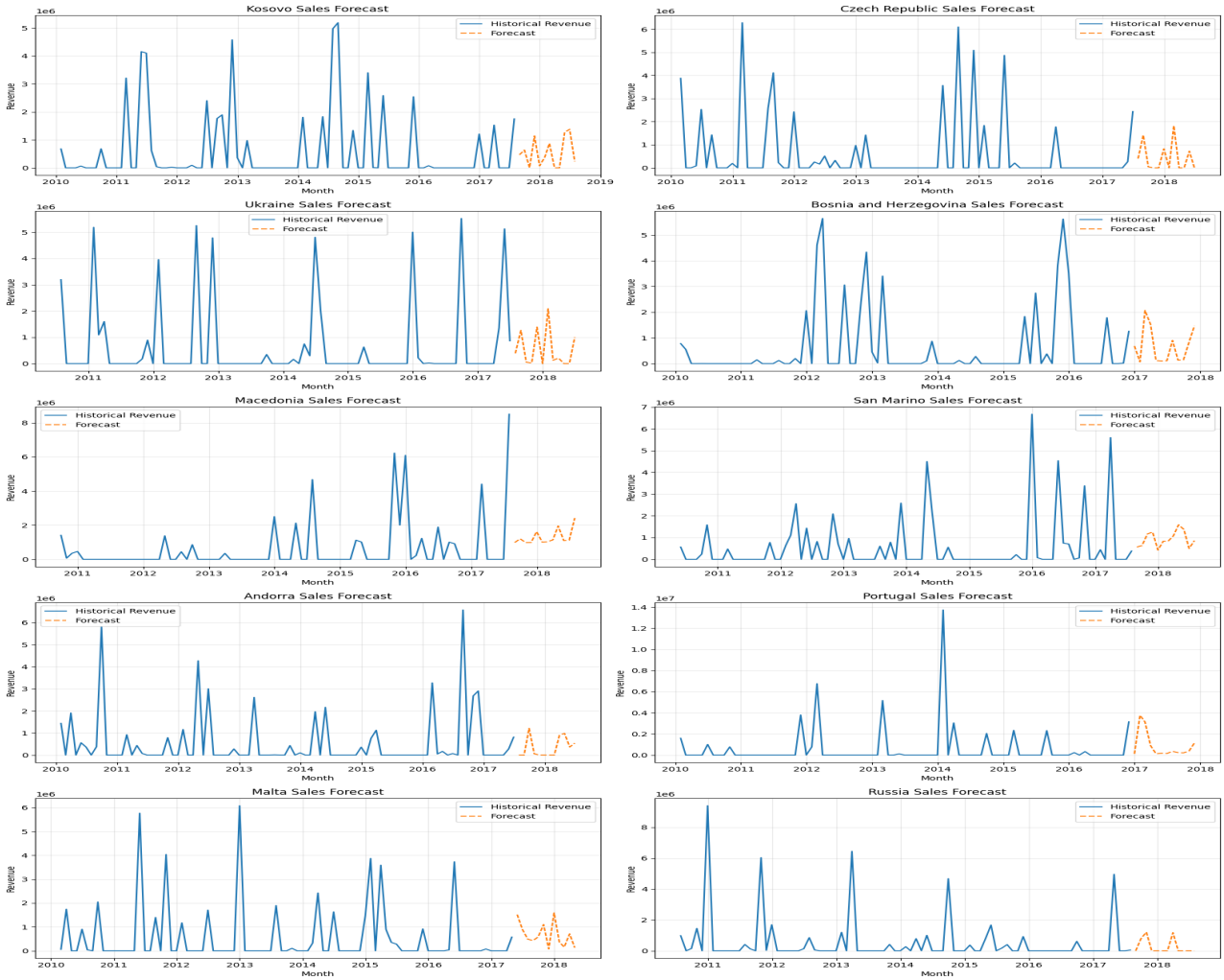
  These are **high-risk, unstable markets** — revenue is not predictable.

- **Small Countries ≠ Small Revenue (Surprising!)**

  Countries like: **San Marino (~47M)**, **Andorra (~47M), Liechtenstein (~29M).** Tiny countries generating huge revenue relative to size due to High-value customers, Niche markets, Tax/business hubs

  Overall Business is not dependent on 1–2 countries, Growth is fragmented and inconsistent.

Country-wise Analysis and Forecast Interpretation



1. **Kosovo**

   Kosovo shows noticeable fluctuations in monthly revenue, with several sharp increases and declines over the observed period. The trend is moderately unstable, indicating that sales are influenced by irregular factors such as promotions or demand variability. Seasonality is present but not strongly consistent. The forecast suggests moderate future growth with continued fluctuations, indicating that while the market is active, it lacks stability.

2. **Czech Republic**

   The Czech Republic demonstrates high variability in sales, with significant peaks followed by sudden drops. This indicates a volatile market environment. The seasonal pattern is weak, and the trend is inconsistent over time. Forecast results show irregular future values, suggesting that predictions should be interpreted cautiously due to the unstable nature of the data.

3. **Ukraine**

   Ukraine exhibits strong spikes in revenue, indicating periods of high demand followed by sharp declines. The trend is not smooth, and seasonal patterns are difficult to identify clearly. The forecast reflects this instability, with large variations expected in future sales. This suggests that external factors heavily influence sales performance.

4. **Bosnia and Herzegovina**

   This country shows relatively strong peaks in certain periods, possibly driven by seasonal demand or promotional campaigns. While some repeating patterns are visible, they are not consistent throughout the series. The forecast indicates continued variability, with occasional high sales periods expected.

5. **Macedonia**

   Macedonia presents one of the most stable patterns among the top countries. The trend is relatively smooth, and sales are consistently maintained over time. Seasonal patterns are more visible compared to other countries. The forecast suggests steady growth with moderate fluctuations, making it one of the more predictable markets in the dataset.

6. **San Marino**

   San Marino shows moderate fluctuations with occasional peaks. The trend is somewhat stable but still affected by irregular changes. Seasonality exists but is not strongly defined. Forecast results indicate moderate growth with periodic variations, suggesting a balanced but slightly unpredictable market.

7. **Andorra**

   Andorra exhibits highly irregular sales patterns with noticeable instability. The data contains significant noise, making it difficult to identify clear trends or seasonality. As a result, the forecast is less reliable and may produce inconsistent values. This indicates that the market is highly volatile.

8. **Portugal**

Portugal shows strong seasonal peaks, suggesting recurring demand patterns, possibly linked to specific months or events. The trend includes both growth and decline phases. The forecast indicates the continuation of these seasonal spikes, making it a market with identifiable cyclical behaviour.

9. **Malta**

Malta demonstrates moderate stability with some recurring patterns. While fluctuations exist, they are less extreme compared to other countries. The forecast suggests relatively stable future performance with occasional increases, indicating a balanced market.

10. **Russia**

Russia shows significant instability with large fluctuations and irregular patterns. The trend is inconsistent, and seasonality is weak. The forecast reflects this volatility, with unpredictable future values. This suggests that the market is influenced by strong external or irregular factors.

Overall Forecast Insights

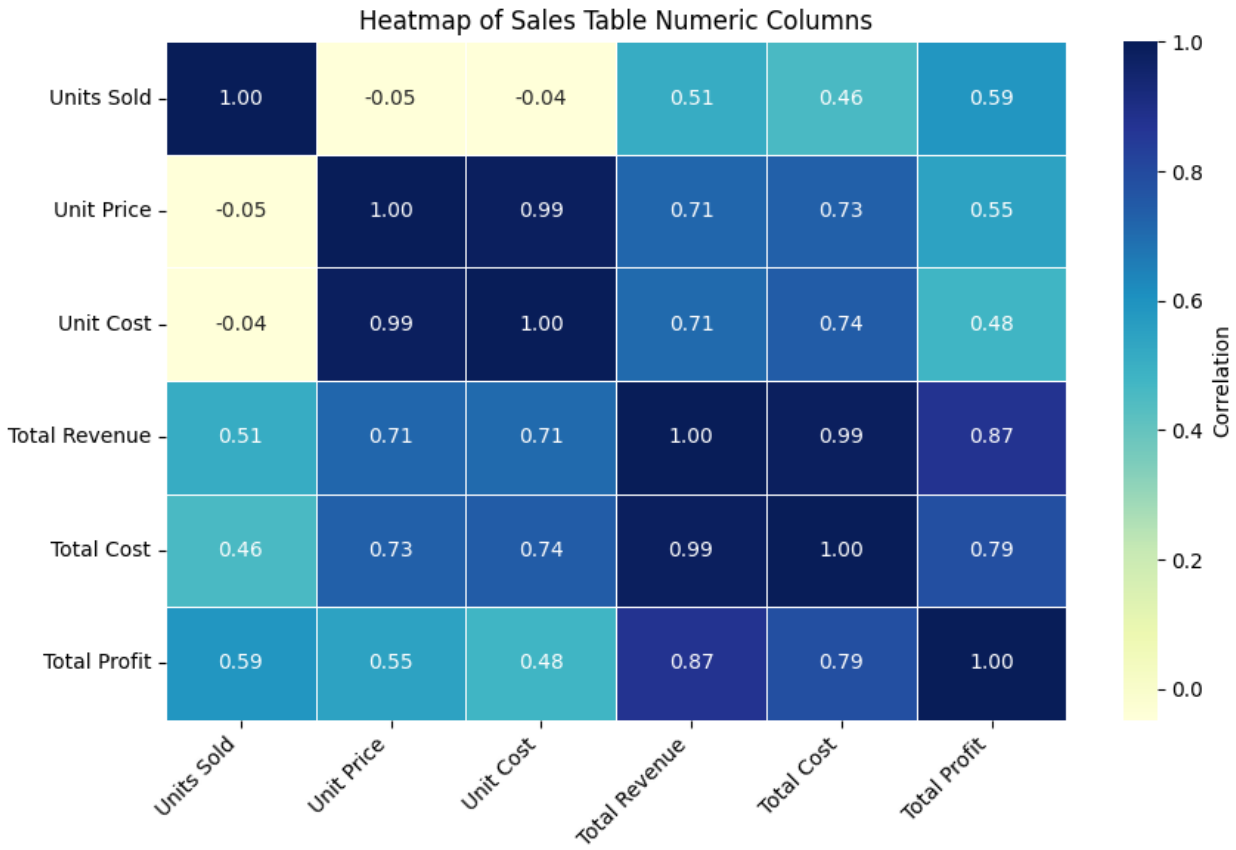
- The majority of countries exhibit **high variability and irregular fluctuations** in sales.
- **Seasonality is weak or inconsistent** in most cases, except for a few countries like Portugal.
- **Trend patterns are non-linear**, indicating periods of growth, decline, and recovery.
- Forecast reliability varies significantly across countries, depending on data stability.
- More stable countries (e.g., Macedonia, Malta) produce **more reliable forecasts**, while volatile countries (e.g., Russia, Andorra) produce **less reliable predictions**.

Final Interpretation

- The analysis highlights that ASDA's sales performance across European markets is heterogeneous. While some countries exhibit stable and predictable behaviour, others are highly volatile. This suggests that forecasting models should be selected carefully based on the characteristics of each country's data rather than applying a single model universally.
- Overall, the results emphasize the importance of understanding data patterns before applying forecasting techniques, as real-world retail data often contains significant noise and irregularities.

Note: For forecast figure see appendices, Forecasting Table for all the Countries

Sales Data Correlation Analysis



Overview

This analysis examines the relationships between key numerical variables, including Units Sold, Unit Price, Unit Cost, Total Revenue, Total Cost, and Total Profit. The goal is to understand the primary drivers of revenue and profitability within the dataset.

Key Findings

Strong Relationship Between Revenue and Cost

One of the most significant observations is the extremely high correlation between total revenue and total cost (0.99). This indicates that costs increase almost proportionally with revenue. The business appears to operate on a model where higher sales volume directly leads to higher expenses.

Revenue as the Main Driver of Profit

Total revenue shows a strong positive relationship with total profit (0.87). This suggests that as revenue increases, profit also increases significantly. However, since cost is also closely tied to revenue, profitability depends on maintaining a balance between revenue growth and cost control.

Pricing Strategy is Closely Linked to Cost

Unit price and unit cost have a near-perfect correlation (0.99). This indicates that pricing is likely based on cost, possibly through a fixed markup strategy. While this ensures consistency, it may limit opportunities to expand profit margins through dynamic pricing.

Moderate Impact of Units Sold

Units sold has a moderate relationship with both revenue (0.51) and profit (0.59). This suggests that increasing sales volume contributes to performance, but it is not the strongest influencing factor compared to pricing and cost.

Low Price Sensitivity in Demand

There is almost no relationship between units sold and unit price (-0.05). This indicates that demand is relatively stable and not significantly affected by price changes. Customers appear to continue purchasing regardless of minor price variations.

Overall Interpretation

The business operates on a price-driven model where revenue and profit are largely influenced by pricing and cost structure. While increasing sales volume contributes to growth, it is not the primary driver. Effective cost management and pricing strategies are critical for improving profitability.

Important Highlights

- Revenue and cost move almost identically, indicating a volume-based cost structure
- Profit growth is strongly dependent on revenue but also influenced by cost efficiency
- Pricing is closely tied to cost, suggesting limited flexibility in margin expansion
- Sales volume has a moderate impact compared to pricing and cost factors
- Demand appears stable and largely unaffected by price changes

Appendices

Forecasting Table for all the Countries

Country	Date	Year	Forecast Revenue
Kosovo	2017-08-31	2017	467714.573
Kosovo	2017-09-30	2017	636030.1567
Kosovo	2017-10-31	2017	6681.16987
Kosovo	2017-11-30	2017	1140644.28
Kosovo	2017-12-31	2017	71401.86984
Kosovo	2018-01-31	2018	384229.5401
Kosovo	2018-02-28	2018	875148.9135
Kosovo	2018-05-31	2018	1269472.403
Kosovo	2018-06-30	2018	1377601.544
Kosovo	2018-07-31	2018	227830.5489
Czech Republic	2017-07-31	2017	401193.2595
Czech Republic	2017-08-31	2017	1416119.237
Czech Republic	2017-09-30	2017	43867.86444
Czech Republic	2017-12-31	2017	824580.5487
Czech Republic	2018-02-28	2018	1824377.011
Czech Republic	2018-05-31	2018	724124.6844
Ukraine	2017-07-31	2017	391407.9862
Ukraine	2017-08-31	2017	1261713.429
Ukraine	2017-09-30	2017	54947.51302
Ukraine	2017-10-31	2017	19022.62076
Ukraine	2017-11-30	2017	1390678.085
Ukraine	2018-01-31	2018	2088219.13
Ukraine	2018-02-28	2018	125602.719
Ukraine	2018-03-31	2018	209469.6638
Ukraine	2018-06-30	2018	1001900.193
Bosnia and Herzegovina	2016-12-31	2016	679362.8294
Bosnia and Herzegovina	2017-01-31	2017	62684.74414
Bosnia and Herzegovina	2017-02-28	2017	2062035.042
Bosnia and Herzegovina	2017-03-31	2017	1515994.379
Bosnia and Herzegovina	2017-04-30	2017	113319.5576
Bosnia and Herzegovina	2017-05-31	2017	91307.93665
Bosnia and Herzegovina	2017-06-30	2017	102582.3081
Bosnia and Herzegovina	2017-07-31	2017	892425.2438
Bosnia and Herzegovina	2017-08-31	2017	121574.6151
Bosnia and Herzegovina	2017-09-30	2017	136767.6289
Bosnia and Herzegovina	2017-10-31	2017	820316.5177
Bosnia and Herzegovina	2017-11-30	2017	1456814.569
Macedonia	2017-08-31	2017	984810.8759
Macedonia	2017-09-30	2017	1202103.225
Macedonia	2017-10-31	2017	979382.1453

Macedonia	2017-11-30	2017	981958.7506
Macedonia	2017-12-31	2017	1621573.894
Macedonia	2018-01-31	2018	1009467.766
Macedonia	2018-02-28	2018	1024089.792
Macedonia	2018-03-31	2018	1135935.376
Macedonia	2018-04-30	2018	1951953.148
Macedonia	2018-05-31	2018	1096209.018
Macedonia	2018-06-30	2018	1118800.522
Macedonia	2018-07-31	2018	2416909.455
San Marino	2017-08-31	2017	563805.8394
San Marino	2017-09-30	2017	634287.2105
San Marino	2017-10-31	2017	1162437.358
San Marino	2017-11-30	2017	1255540.108
San Marino	2017-12-31	2017	425499.4456
San Marino	2018-01-31	2018	831139.857
San Marino	2018-02-28	2018	830340.6516
San Marino	2018-03-31	2018	1078540.598
San Marino	2018-04-30	2018	1588597.282
San Marino	2018-05-31	2018	1370502.437
San Marino	2018-06-30	2018	496893.7594
San Marino	2018-07-31	2018	859679.6526
Andorra	2017-09-30	2017	1209741.529
Andorra	2017-10-31	2017	58524.29096
Andorra	2018-03-31	2018	901319.3236
Andorra	2018-04-30	2018	984087.351
Andorra	2018-05-31	2018	362271.7417
Andorra	2018-06-30	2018	541719.9122
Portugal	2016-12-31	2016	119307.0022
Portugal	2017-01-31	2017	3764292.814
Portugal	2017-02-28	2017	3124844.897
Portugal	2017-03-31	2017	892466.8902
Portugal	2017-04-30	2017	125570.3294
Portugal	2017-05-31	2017	170086.8314
Portugal	2017-06-30	2017	148892.7517
Portugal	2017-07-31	2017	351588.9329
Portugal	2017-08-31	2017	232995.4131
Portugal	2017-09-30	2017	208824.4328
Portugal	2017-10-31	2017	378086.6239
Portugal	2017-11-30	2017	1140824.938
Malta	2017-05-31	2017	1519536.378
Malta	2017-06-30	2017	898938.8698
Malta	2017-07-31	2017	489059.3367
Malta	2017-08-31	2017	428616.0003
Malta	2017-09-30	2017	585169.338
Malta	2017-10-31	2017	1104041.106

Malta	2017-11-30	2017	77249.86056
Malta	2017-12-31	2017	1590660.819
Malta	2018-01-31	2018	366295.6032
Malta	2018-02-28	2018	156795.9023
Malta	2018-03-31	2018	711997.7621
Malta	2018-04-30	2018	130309.7576
Russia	2017-09-30	2017	802456.8428
Russia	2017-10-31	2017	1224624.021
Russia	2018-03-31	2018	1192925.237

Asda Total Yearly Sales 2010-2017

Row Labels	Sum of Total Revenue
2010	**$20,93,82,247.40**
2011	**$23,19,37,556.67**
2012	**$28,54,85,891.76**
2013	**$18,29,62,359.28**
2014	**$23,77,78,813.23**
2015	**$22,79,54,098.84**
2016	**$19,15,20,956.77**
2017	**$13,66,00,474.16**
Grand Total	**1703622398**

Asda Total Monthly Sales 2010-2017

Row Labels	Sum of Total Revenue
Jan	$15,79,03,881.67
Feb	$14,15,44,945.25
Mar	$17,06,92,280.05
Apr	$11,48,07,392.50
May	$15,77,28,838.17
Jun	$16,60,83,474.31
Jul	$15,59,47,385.47
Aug	$11,72,05,903.92
Sep	$12,73,24,498.78
Oct	$10,78,32,960.98
Nov	$13,94,55,445.82
Dec	$14,70,95,391.19
Grand Total	**$1,70,36,22,398.11**